Sprint 1 Proposal

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1. Project idea

This on-demand marketplace web application helps customers discover and book barbers near them. Customers browse shops by location (e.g. Clapham, Brixton, Balham), preferred haircut or service, and price, then view availability and request an appointment. Barbers and shop owners manage their profile, services, prices and time slots.

Business type: a local marketplace connecting independent barber shops with customers.

2. Problem statement

People often struggle to find a barber who is nearby, offers the style they want and fits their budget. Information is scattered across Instagram, Google Maps and word of mouth, and checking availability usually means calling or messaging. Meanwhile, barbers lose income to empty slots and no-shows, and have no simple way to publish their services and availability. Customers also need a way to message barbers directly, share hairstyle ideas in a community space, and read ratings and reviews before choosing.

Customers can benefit from real-time messaging as a convenient way to message barbers directly to clarify haircut requirements, discuss appointment times, or ask about prices before booking. A community space for sharing hairstyle ideas, recommendations, and experiences would also be useful, along with a rating and review system to help customers choose the right barber.

How might we...

- ...help customers find a barber who matches their area, style and budget in under a minute?
- ...let customers see real availability without phoning around?
- ...help barbers fill empty slots and manage bookings with less admin?
- ...make prices and services transparent before a customer visits?

3. Purpose of the application

1. Let customers search and filter barber shops by area, style, price and rating.
2. Let customers view availability and book, change or cancel appointments.
3. Let barbers and owners manage their shop profile, services, prices and time slots.
4. Give barbers simple insights (most booked services, busiest times, unfilled slots) to support business decisions.
5. Enable direct messaging between customers and barbers to clarify services, arrange appointments, and negotiate prices where applicable.
6. Provide a community forum for sharing hairstyle inspiration, asking for recommendations, and discussing barber experiences.
7. Allow customers to rate barbershops and leave written reviews after completed appointments.

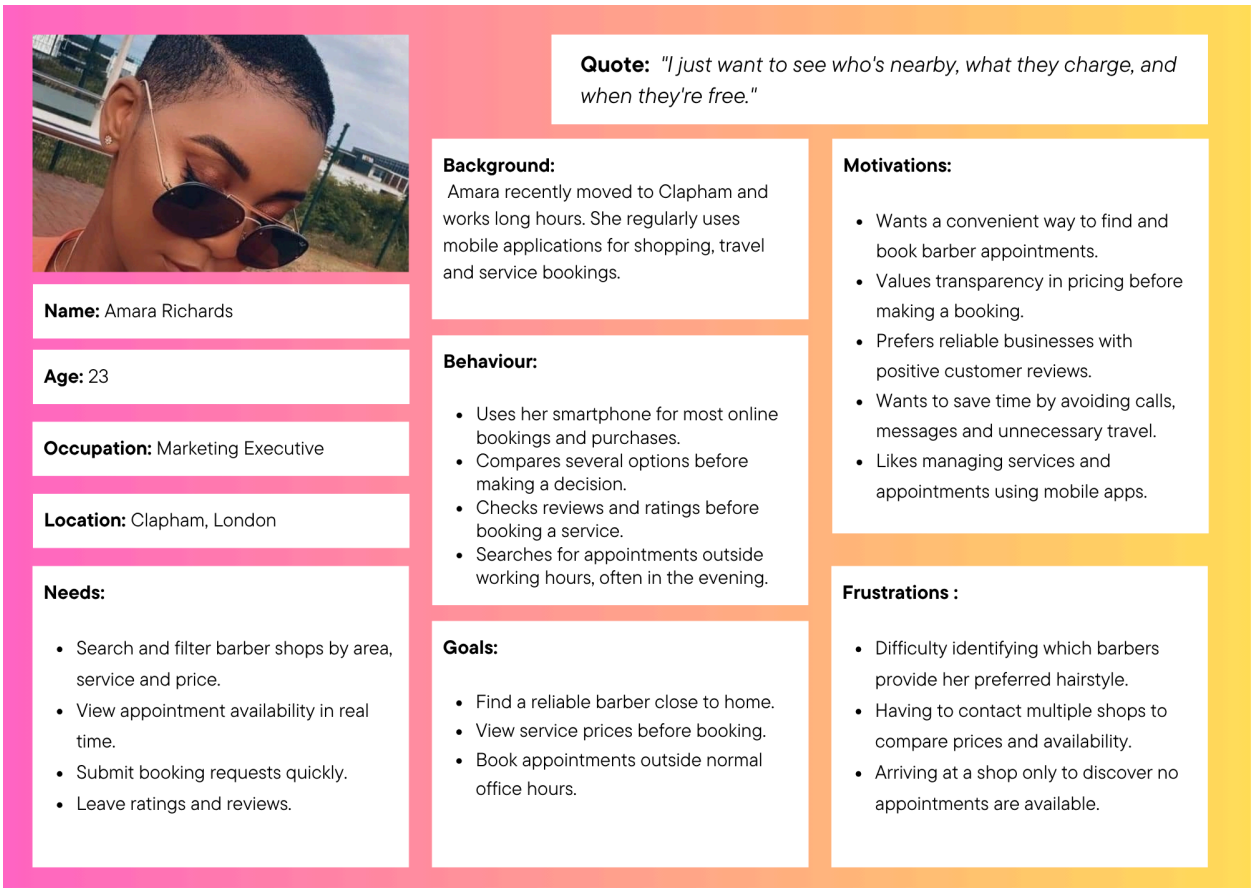
This application is well suited to CRUD operations. For example:

Entity	Customer actions (role)	Barber / owner actions (role)	System / admin operations
Barber shops	Read and search shop profiles and ratings	Create profile, update details, delete shop listing	Index shops by location
Services and pricing	Read services and view price lists	Create services, update prices and duration, delete old services	Recalculate price ranges
Availability and slots	Read free slots in real time	Create work hours, update break times, delete or block slots	Auto-expire past time slots
Appointments	Create booking, read history, update time, delete or cancel	Read schedule, update status (accepted, declined or completed)	Send booking confirmations
Reviews and analytics	Create review, read shop ratings	Read insights (busiest times, top services)	Aggregate rating scores and slot statistics
Messages	Create, read and reply to messages	Read and reply to customer enquiries	Store messages and enforce access permissions
Community forum	Create posts, comment and read discussions	Participate in discussions and share advice	Moderate reported content

4. Business questions

1. Which barber shops offer the services a customer is looking for within their preferred area?
2. Which barber shops have available appointment slots on a customer's preferred date and time?
3. What services does each barber shop offer, and how much does each one cost?
4. Which services are booked most frequently, and in which areas are they most popular?
5. Which days and times see the highest demand for appointments, and which time slots are least utilised?
6. Which barber shops have the highest customer ratings and the most repeat customers?
7. How often do enquiries via messaging lead to confirmed bookings?

5. Personas



Name: Amara Richards

Age: 23

Occupation: Marketing Executive

Location: Clapham, London

Needs:

- Search and filter barber shops by area, service and price.
- View appointment availability in real time.
- Submit booking requests quickly.
- Leave ratings and reviews.

Quote: *"I just want to see who's nearby, what they charge, and when they're free."*

Background:
Amara recently moved to Clapham and works long hours. She regularly uses mobile applications for shopping, travel and service bookings.

Behaviour:

- Uses her smartphone for most online bookings and purchases.
- Compares several options before making a decision.
- Checks reviews and ratings before booking a service.
- Searches for appointments outside working hours, often in the evening.

Goals:

- Find a reliable barber close to home.
- View service prices before booking.
- Book appointments outside normal office hours.

Motivations:

- Wants a convenient way to find and book barber appointments.
- Values transparency in pricing before making a booking.
- Prefers reliable businesses with positive customer reviews.
- Wants to save time by avoiding calls, messages and unnecessary travel.
- Likes managing services and appointments using mobile apps.

Frustrations :

- Difficulty identifying which barbers provide her preferred hairstyle.
- Having to contact multiple shops to compare prices and availability.
- Arriving at a shop only to discover no appointments are available.



Quote: *"If my diary managed itself, I could focus on cutting hair."*

Name: Marcus Raymond

Age: 34

Occupation: Barber Shop Owner

Location: Balham, London

Needs:

- Publish services, prices and availability.
- Accept or decline appointment requests.
- Manage bookings from a single platform.
- View booking trends and service popularity.

Background:
Marcus owns a two-chair barber shop and currently relies on walk-in customers, phone calls and social media to manage bookings.

Behaviour:

- Updates his schedule regularly as appointments change.
- Uses social media to advertise services and attract customers.
- Responds to booking enquiries between appointments.
- Adjusts availability and promotions based on customer demand.

Goals:

- Increase bookings during quieter periods.
- Reduce no-shows and scheduling conflicts.
- Attract more local customers.

Motivations:

- Wants to increase bookings and attract new customers.
- Aims to maximise appointment capacity, particularly during quiet periods.
- Wants a simpler way to manage bookings and availability.
- Seeks to reduce no-shows and scheduling conflicts.
- Wants insights into customer demand and service popularity to improve business performance.

Frustrations :

- Frequent interruptions from booking enquiries.
- Double-bookings caused by manual scheduling.
- Lack of information about which services generate the most revenue.

6. Task definition and initial backlog

To deliver the minimum viable product (MVP), development is divided into tasks across three core phases.

Phase 1: Platform setup and data architecture

- Set up the Git repository, branching strategy and Kanban task board.
- Design the relational database schema (Users, Shops, Services, Availability Slots, Bookings, Ratings, Messages, Forum Posts and Comments).
- Implement authentication and user session handling (Customer and Barber roles).

Phase 2: Core business and CRUD features

- Build the Barber Dashboard to create, update and delete shop profiles, service menus, pricing, work hours and time slots.
- Build the Customer Discovery Interface with filtering by area, price and service type.

- Implement real-time availability scheduling and the appointment creation, rescheduling and cancellation workflow.
- Build booking management views for barbers to accept, decline or update booking status.
- Send booking confirmation notifications.
- Implement direct messaging between customers and barbers, with access permissions.
- Allow customers to submit ratings and written reviews for completed appointments.
- Display average ratings and customer reviews on barber profiles.

Phase 3: Insights, community and quality assurance

- Build a shop analytics panel querying most booked services, peak booking hours and unbooked slots.
- Build a community forum with posts, comments and moderation of reported content.
- Test messaging permissions, review submissions and forum moderation.
- Carry out end-to-end integration testing and UI responsiveness checks.

7. Code of conduct

7.1 Expected behaviour

Be respectful: All group members should treat one another with courtesy, professionalism and respect. Communication should remain constructive, whether in person, online or through messaging platforms, and should avoid offensive, discriminatory or inappropriate language.

Understand different perspectives: Group members come from diverse backgrounds and may have different experiences, opinions and working styles. Everyone should listen actively, consider alternative viewpoints and be open to compromise when making decisions. Constructive disagreement is encouraged but should remain respectful and focused on the project.

Communicate responsibly: Members should communicate clearly and promptly regarding meetings, deadlines and project updates. If a member is unable to complete a task or attend a meeting, they should inform the group as soon as possible.

Contribute fairly: Each member is expected to contribute equally by completing assigned tasks on time and participating in discussions, meetings and collaborative activities.

Maintain academic integrity: All submitted work must be original or appropriately referenced. Members should avoid plagiarism, fabrication of information, or presenting another person's work as their own.

Git and task board standards: Every team member must actively manage their assigned tasks on the project board and commit code regularly. Bulk commits by a single developer are prohibited.

Communication protocol: Queries about blockers in the group chat must be acknowledged within 24 hours. A member who anticipates missing a sprint milestone must give at least 48 hours' notice.

Code reviews: All merges into the main branch require review or approval from at least one other team member.

7.2 Behaviour that will not be tolerated

Violence and threats of violence: Any form of physical violence, intimidation, threats, harassment or behaviour that makes others feel unsafe is strictly prohibited.

Personal attacks: Insults, name-calling, bullying, discriminatory remarks, harassment or targeting individuals based on personal characteristics will not be tolerated.

Disruptive behaviour: Repeated interruptions, refusal to cooperate, ignoring agreed group decisions, missing meetings without explanation, consistently failing to complete assigned tasks, or any behaviour that negatively affects the group's progress is unacceptable.

7.3 Consequences of unacceptable behaviour

If a group member engages in unacceptable behaviour, the following actions may be taken, depending on the severity of the issue:

- An informal discussion to address the concern and seek resolution.
- A formal warning from the group.
- Documentation of repeated issues and communication with the module lecturer or tutor.
- Reassignment of responsibilities within the group where appropriate.
- Reporting of serious misconduct to the university in accordance with institutional policies.

7.4 Reporting

Any group member who experiences or witnesses behaviour that violates this Code of Conduct should report it to the group as soon as possible. Reports should be treated confidentially and respectfully.

Where an issue cannot be resolved within the group, members should raise it with the module lecturer, tutor or relevant university staff member. Retaliation against anyone who reports concerns in good faith will not be tolerated.

8. Project links

- **Task board:** https://trello.com/invite/b/6ac4fbb57b91466f6166b9f5/ATTIe2076944628d39e384395f9acd707e8fA7395338/🧩-group-project-workflow-board
- **GitHub repository:** https://github.com/johnsontom/barber-platform

9. Ethical considerations

As a digital marketplace connecting local businesses with consumers, the platform handles personal data, transactional workflows and reputation metrics. To ensure responsible and legally compliant software development, the project will address the following ethical dimensions:

- **Data privacy and protection (GDPR compliance):** Because the platform processes user identities, contact details and booking records, all handling of personal data must comply with the UK GDPR. Article 5 establishes core obligations, including lawful processing, transparency, purpose limitation and strict data minimisation [1]. Accordingly, the system will collect only the personal information essential to core marketplace functions. Security controls, such as strong

password hashing, restricted role-based access and safeguards against unauthorised data entry, will be implemented in line with Article 32 [1]. Privacy-by-design principles, mandated under Article 25, will guide default system configurations [1], and users retain the right to request erasure of their data under Article 17 where legally applicable [1].

- **Algorithmic neutrality and recommendation logic:** To support fair vendor discovery, the platform's ranking and recommendation mechanisms must minimise systemic bias and make clear how listings are ordered. New or independent barbers may have limited historical feedback, which can unintentionally reduce their exposure. The system should therefore monitor how ranking criteria affect merchant visibility and implement safeguards that prevent disproportionate disadvantage for recent entrants. Scholarship highlights the need to distinguish statistical optimisation from broader regulatory compliance obligations [2], while the transparency requirements of Article 5(1)(a) of the UK GDPR remain central to responsible algorithmic processing [1].
- **Digital accessibility and inclusion:** To accommodate users of varying digital literacy and physical abilities, the customer discovery and booking interfaces will adhere to the Web Content Accessibility Guidelines (WCAG), ensuring universal usability [3].

References

[1] UK General Data Protection Regulation, Regulation (EU) 2016/679, arts. 5, 17, 25 and 32. [Online]. Available: https://www.legislation.gov.uk/eur/2016/679/contents. [Accessed: Oct. 12, 2026].

[2] J. Liu, X. Luo and G. H. Tang, "Bridging fairness in machine learning: A legal critique of technical strategies," *Computer Law & Security Review*, vol. 61, p. 106339, May 2026, doi: 10.1016/j.clsr.2026.106339.

[3] World Wide Web Consortium (W3C), "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Dec. 12, 2024. [Online]. Available: https://www.w3.org/TR/WCAG22/. [Accessed: Oct. 12, 2026].